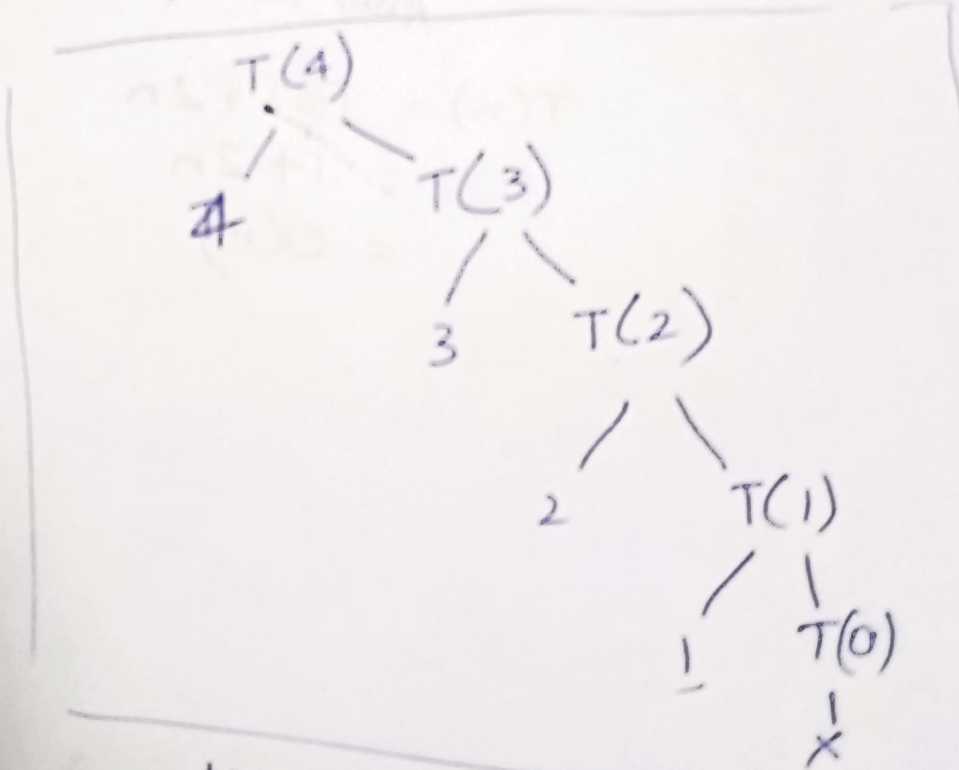
①

```
void Test(int n) — T(n)
{
  if(n>0) —— ①
  {
    Print("%d",n); —①
    Test(n-1); — T(n-1)
  }
}
```

$$T(n) = T(n-1) + 2$$

$$\begin{cases} T(0) = 1 & n = 0 \\ T(n) = T(n-1) + 2 & n > 0 \end{cases}$$

n = 4



function execution = 5 = n+1

Print = 4 = n

O(n)

$$T(n) = T(n-1) + 2 \quad \text{— (1)}$$

$$T(n-1) = T(n-2) + 2$$

$$T(n) = T(n-2) + 2 + 2 \quad \text{— (II)}$$

$$T(n-2) = T(n-3) + 2$$

$$T(n) = T(n-3) + 2 + 2 + 2$$

$$T(n) = T(n-3) + 3 \times 2$$

$$\downarrow k$$

$$T(n) = T(n-k) + 2k$$

Assume $k = n$

$$T(n) = T(0) + 2n$$
$$= 1 + 2n$$
$$= O(n)$$

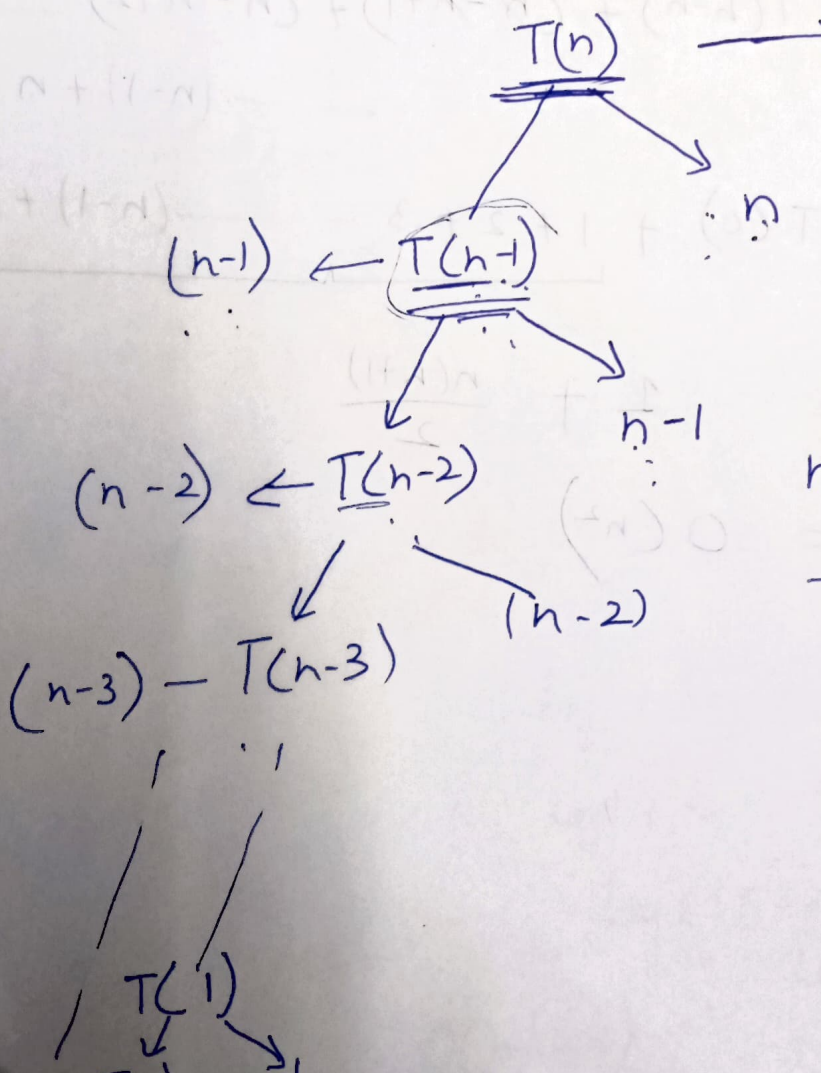
```
void Test (int n)          — T(n)
{
  if (n>0)                 ——— 1
  {
    for (i=0; i<n; i++)    — n+1
    {
      Printf ("%d", n);    — n
    }
    Test(n-1);             — T(n-1)
  }
}
```

$$T(n) = \begin{cases} 1 & n=0 \\ T(n-1)+n & n>0 \end{cases}$$

$$T(n) = T(n-1) + n+1 + n + 1$$

$$= T(n-1) + 2n + 2$$

$$T(n) = T(n-1) + 2n \Rightarrow T(n-1)+n$$



$n + (n-1) + (n-2)$

$- - - \; 2 + 1$

$$\frac{n(n+1)}{2} = O(n^2)$$

$$T(n) = T(n-1) + n$$

$$T(n-1) = T(n-2) + (n-1)$$

$$T(n) = T(n-2) + (n-1) + n$$

$$T(n-2) = T(n-3) + (n-2)$$

$$Tn = T(n-3) + (n-2) + (n-1) + n$$

in terms of K

$$Tn = T(n-K) + (n-(K-1)) + (n-(K-2))$$

$$\text{---} \quad (n-1)+n$$

Assume $K = n$

$$T(n) = T(n-n) + (n-K+1) + (n-n+2) \text{ ---}$$

$$\text{---} \quad (n-1)+n$$

$$T(n) = T(0) + \underline{1+2+3 \text{ ---} \text{---} (n-1)+n}$$

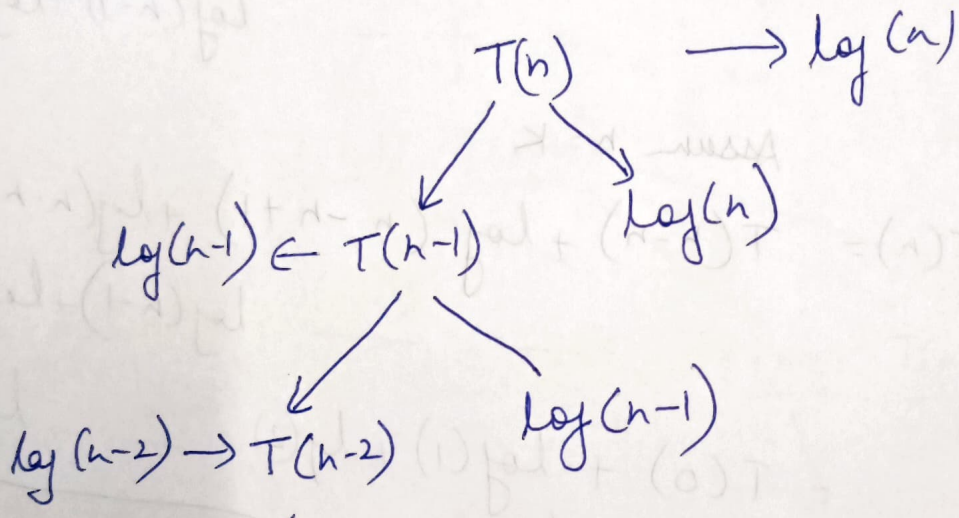
$$= 1 + \frac{n(n+1)}{2}$$

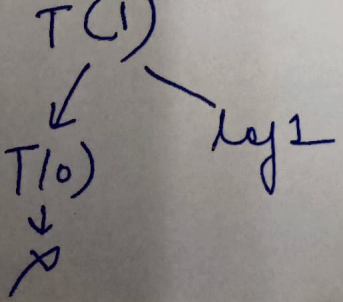
$$= O(n^2)$$

```
d Test(int n)  —T(n)
{
  if(n>0)  — ①
  {
    for(i=1; i<n; i=i*2) →log(n)+1
    {
      Printf("%d", i);  → log(n)
    }
    Test(n-1);  → T(n-1)
  }
```

$$T(n) = T(n-1) + 2\log(n) + 2$$

$$T(n) = T(n-1) + \log(n)$$





$\log 1 + \log 2 + \log 3 \cdots \log n$

$\log 1 + \log 2 + \log 3 \cdots \log(n-1) < \log n$

$\log(1 \times 2 \times 3 \cdots n-1 \times n)$

$\log(n!)$ $(n! > n^n)$

$O(n \log n)$ $\log n^n \Rightarrow n \log n$

$T(n) = T(n-1) + \log n$

$T(n-1) = T(n-2) + \log(n-1)$

$T(n) = T(n-2) + \log(n-1) + \log n$

$T(n-2) = T(n-3) + \log(n-2)$

$T(n) = T(n-3) + \log(n-2) + \log(n-1) + \log n$

for k

$T(n) = T(n-k) + \log(n-(k-1)) + \log(n-(k-2))$

$\log(n-1) + \log n$

Assum $n = k$

$T(n) = T(n-n) + \log(n-n+1) + \log(n-n+2)$

$\log(n-1) + \log n$

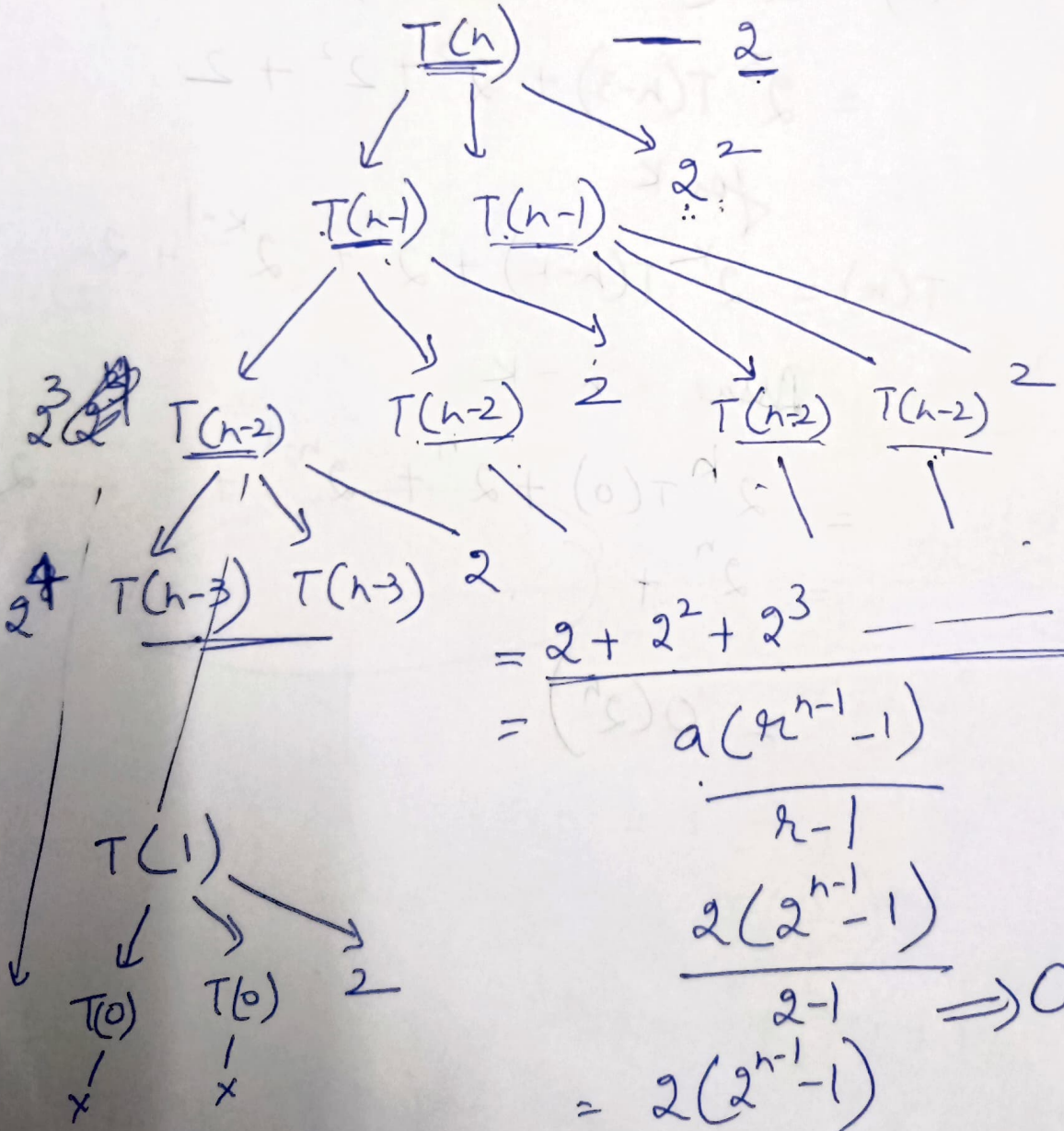
$= T(0) + \log(1) + \log(2) \quad \log(n)$

$= O(n \log n)$

$T(n) = 2T(n-1) + 1$

```
Algorithm Test(int n)  — T(n)
{
  if(n>0)  — (1)
  {
    Printf("%d", n);  — (1)
    Test(n-1);  — T(n-1)
    Test(n-1);  — T(n-1)
  }
}
```

$T(n) = 2T(n-1) + 2$



$= 2 + 2^2 + 2^3 \text{ ---- } 2^n$

$= \frac{a(r^{n-1} - 1)}{r-1}$

$\frac{2(2^{n-1} - 1)}{2-1} \Rightarrow O(2^n)$

$= 2(2^{n-1} - 1)$

$$T(n) = 2T(n-1) + 2$$

$$T(n-1) = 2T(n-2) + 2$$

$$T(n) = 2(2T(n-2) + 2) + 2$$

$$= 2^2 T(n-2) + 2^2 + 2$$

$$T(n-2) = 2T(n-3) + 2$$

$$T(n) = 2^2(2T(n-3) + 2) + 2^2 + 2$$

$$= 2^3 T(n-3) + 2^3 + 2^2 + 2$$

for K

$$T(n) = 2^K T(n-K) + 2^K + 2^{K-1} + 2 - -$$

Assume $n = K$

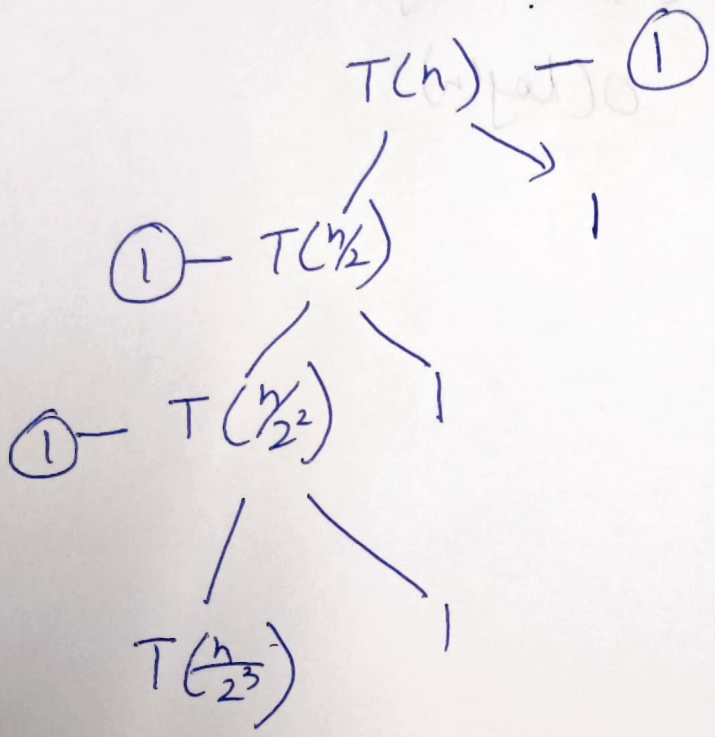
$$= 2^n T(0) + 2^n + 2^{n-1} + - - - 2$$

$$= 2^n + \cdots$$

$$= O(2^n)$$

```
Test(int n)
{
  if (n>1)
  {
    Printf("%d", n);
    Test(n/2)
  }
}
```

$$T(n) = T(n/2) + 1$$



$$\frac{n}{2^k} = 1$$

$$2^k = n$$

$$k = \log_2 n$$

$O(\log_2 n)$

$$T(n) = T(n/2) + 1$$

$$T(n/2) = T\left(\frac{n}{2^2}\right) + 1$$

$$T(n) = T\left(\frac{n}{2^2}\right) + 1 + 1$$

$$= T\left(\frac{n}{2^2}\right) + 2$$

$$T\left(\frac{n}{2^2}\right) = T\left(\frac{n}{2^3}\right) + 1$$

$$T(n) = T\left(\frac{n}{2^3}\right) + 3$$

$$T(n) = T\left(\frac{n}{2^k}\right) + k$$

$$\frac{n}{2^k} = 1$$

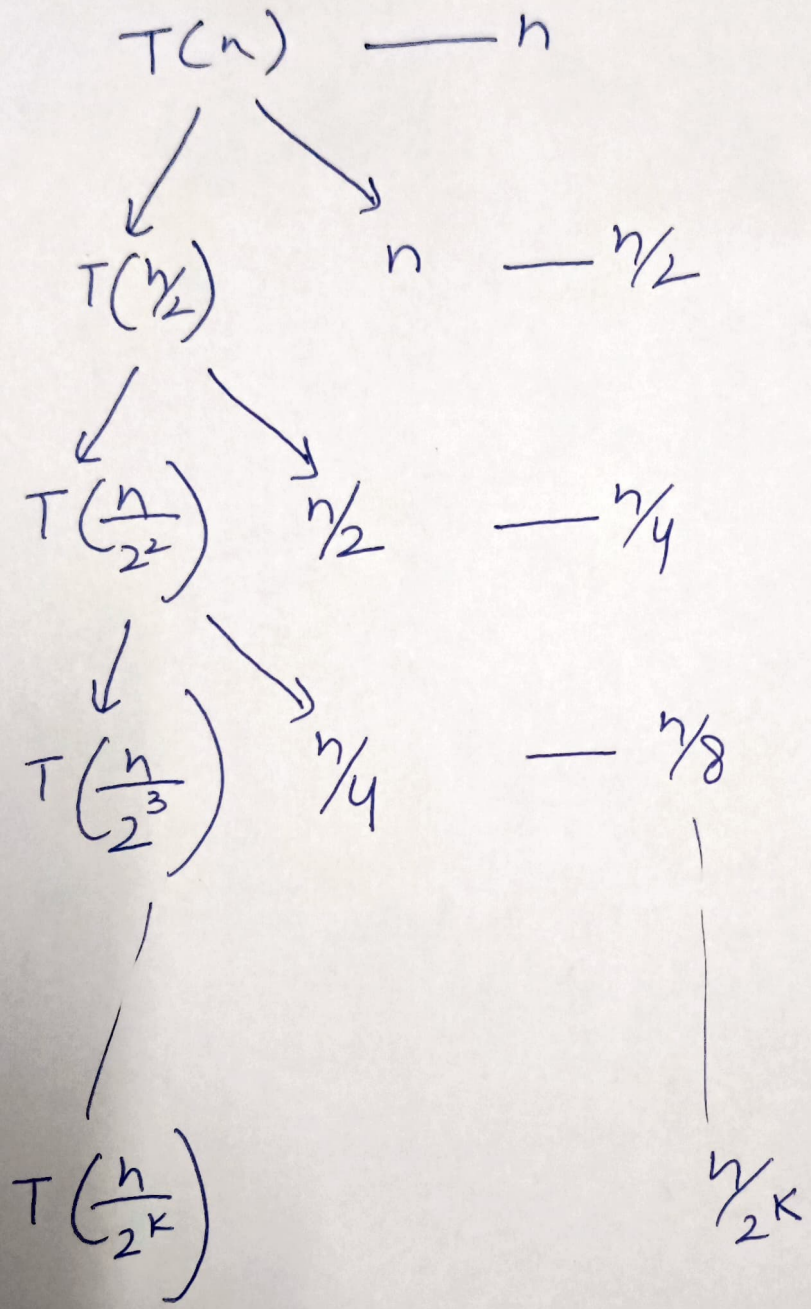
$$2^k = n$$

$$k = \log n$$

$$= T(1) + \log n$$

$$= O(\log(n))$$

$$T(n) = \begin{cases} 1 & n=1 \\ T(n/2)+n & n>1 \end{cases}$$



$$= n + \frac{n}{2} + \frac{n}{4} \cdots \cdots \frac{n}{2^k}$$

$$= n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots \frac{1}{2^k}\right)$$

$$= n \sum_{i=0}^{k} \frac{1}{2^k} = O(n)$$